- Skewness describes the degree to which a distribution is asymmetric about its mean. An asset return distribution with positive skewness has frequent small losses and a few extreme gains compared to a normal distribution. An asset return distribution with negative skewness has frequent small gains and a few extreme losses compared to a normal distribution. Zero skewness indicates a symmetric distribution of returns.
- Kurtosis measures the combined weight of the tails of a distribution relative to the rest of the distribution. A distribution with fatter tails than the normal distribution is referred to as fat-tailed (leptokurtic); a distribution with thinner tails than the normal distribution is referred to as thin-tailed (platykurtic). The kurtosis of a normal distribution is 3.
- The correlation coefficient measures the association between two variables. It is the ratio of covariance to the product of the two variables' standard deviations. A positive correlation coefficient indicates that the two variables tend to move together, whereas a negative coefficient indicates that the two variables tend to move in opposite directions. Correlation does not imply causation, simply association. Issues that arise in evaluating correlation include the presence of outliers and spurious correlation.

MEASURES OF CENTRAL TENDENCY AND LOCATION 2

☐ calculate, interpret, and evaluate measures of central tendency and location to address an investment problem

In this lesson, our focus is on measures of central tendency and other measures of location. A **measure of central tendency** specifies where the data are centered. For a return series, a measure of central tendency shows where the empirical distribution of returns is centered, essentially a measure of the "expected" return based on the observed sample. **Measures of location**, mean, the **median**, and the **mode** include not only measures of central tendency but also other measures that illustrate other aspects of the location or distribution of the data.

Frequency distributions, histograms, and contingency tables provide a convenient way to summarize a series of observations on an asset's returns as a first step toward describing the data. For example, a histogram for the frequency distribution of the daily returns for the fictitious EAA Equity Index over the past five years is shown in Exhibit 1.